FlexiMetal® 35 Metallic Paste

Deep Materials

Product Description

FlexiMetal is a patented, metallic thermal interface material (TIM) with very high thermal conductivity. It sits between a heat source and a heat sink and gives heat an ultra-low-resistance path out. It works under the lid as TIM1, on a bare die as TIM1.5, or on top of the lid as TIM2.

FlexiMetal is not liquid metal. Pure gallium and eutectic gallium alloys such as Ga-In and Ga-In-Sn are liquids with very low viscosity. FlexiMetal, in contrast, is a colloidal suspension of a carbon allotrope in a liquid-metal alloy. It behaves as a highly viscous, flexible paste and contains no organic material. Its shear-thinning rheology lets it flow under assembly pressure to conform to die warpage and cold-plate flatness, then hold its shape at rest.

That high viscosity keeps the material in place and greatly reduces the risk of migration. Unlike other liquid-metal TIMs, FlexiMetal does not need a containment dam, and it also resists oxidation better than typical liquid-metal TIMs.

Typical Applications

- Laptops
- Server CPUs and GPUs
- Networking switch processors
- Smartphones
- Under the lid (TIM1) applications
- Power electronics

Main Features

- FlexiMetal is not liquid metal
- Ultra-low thermal resistance even at thicker BLTs needed for die warpage/flatness
- Stable up to 250 °C; boiling point >2200 °C
- Diffusion-bonds to copper (plated or bare) and bare silicon — no backside metallization needed
- Improves transient, not just steady-state, thermal performance by moving heat quickly from silicon to module
- Validated to 1,000+ hrs at 85°C/85% RH
- Can be factory-bonded to heat sink/lid via patent-pending process to eliminate interfacial resistance — contact Deep Materials

Application Tips

- Keep the product away from fire and boiling water
- Make sure the application interface is clean before use
- Can be screen-printed at >130 um thickness
- FlexiMetal is safe to touch

Storage and Expiration Date

- The product should be stored at room temperature
- Shelf life is 2 years from the date of manufacture

Property & Units	Values
Thermal conductivity (W/m°K)	35
Density (g/ml)	4.4
Appearance	Silvery
Viscosity at Shear rate of 10/s (Pa-s)	~ 100
Young's Modulus (25 °C – material in solid state) (MPa)	0.25
Melting onset (°C)	~ 7
Boiling point (°C)	>2200
Thermal Impedance, BLT ~ 130µm; (°C-cm^2/W)	~ 0.037

Note: The data in this table is preliminary in nature and is subject to change. FlexiMetal technology still in development. This product is subject to pending patents in the US and worldwide.

The data provided are typical values derived from laboratory testing and are intended for reference only. They should not be used as specifications or as a basis for design without independent validation.